

Learning Natural Language Processing (NLP) is learn about how computer understands, interprets and generate human language. It is a combination between computer science and linguistic to process text to speech, translators and a chatbots. During classes, steps of NLP pipelines include text preprocessing, tokenization, lemmatization, stemming, etc. are taught to deepen knowledge in NLP and the Midterm test here make it's more applicable when it requires students to apply all the knowledge that were taught to a real-life problem. NewsBot Intelligence System describes a general category of automated system that use artificial intelligence and natural language processing to gather, process, summarize and distribute news content.

The dataset "BBC News Train" is used in this assignment with five categories: sports, business, entertainment, politics and tech in 1490 total articles. This project incorporated multiple natural language processing techniques, including tokenization, TF-IDF vectorization, part-of-speech tagging, and named entity recognition (NER) using spaCy. Integrating outputs from POS tagging, sentiment analysis, and entity extraction led to DataFrame mismatches and missing feature values, causing model training errors. I addressed memory limitations for large feature matrices by adopting sparse matrices and efficient data structures. Ensuring visualization compatibility (dependency parsing, confusion matrices) between Matplotlib, Seaborn, and Jupyter Notebook rendering was also a key challenge that ultimately deepened my technical proficiency.

Through this project, I gained a clearer understanding of using AI in business field like categorizing news. A huge amount of information is collected every day and it could take several hours for a human to categorize and collect useful information in a traditional way. Using NewsBot Intelligence System makes it simpler to group articles in the same category together. In this project, it has the ability to classify articles into categories such as business, entertainment, sports, politics and tech which makes it a strong commercial application. The models offer news agencies, marketing platforms, and aggregators a powerful tool for better organizing content, enhancing recommendations, and enabling precise ad targeting. However, the Naïve Bayes confusion matrix only has moderate accuracy when it confuses

between business and tech articles. This is probably because of the words that both share in these two articles like Apple, giant tech, Microsoft, etc. Although with some misclassification between these two articles, the classifier could still flag relevant market-related news for financial analytics. Looking ahead, improving the model's accuracy is the direct path to cutting manual effort, maximizing efficiency, and supporting the real-time analytical demands of media clients.

For better use of this training, there are some directions in which this project can be improved to achieve greater efficiency and performance. Deep learning integration should be used to enhance contextual understanding and improve classification accuracy. For example, BERT is a language model using a deep learning called Transformer. It can process entire sentence from left-to-right and right-to-left directions which allows better understand the context of words. Furthermore, combining NER with sentimental analysis makes a deeper level of interpretation by linking specific entities such as people, organizations or locations.

Throughout this project, I gain hands-on experience when I can apply the theories that I have learned to train a real-world project by classifying articles into suitable categories using TF-IDF, NER, etc. combines with sentimental analysis makes it possible to interpret if the article is negative, positive or neutral. Machine learning training, natural language processing and data visualization are important skills relating to Artificial Intelligence are also applied in this project. I also developed a deeper understanding of how algorithms like Naive Bayes operate mathematically and practically in real-world datasets. I gained confidence in using advanced Python libraries such as scikit-learn, spaCy, matplotlib, and seaborn, as well as in interpreting complex outputs like confusion matrices, feature importance coefficients, and entity recognition patterns. Working this project on my own helps me improve my time management and how to use AI assistant to help me understand more about this project but do not rely on it. This experience strengthens not only my knowledge in NewsBot Intelligence system but also how to be discipline and patient to troubleshoot the errors and unexpected results.